1. Scope & Planning

STEP 1 - Scope & Planning | Injection Molding - PEEK Lumbar Interbody Fusion Cage

Field	Detail
Process Name	Injection Molding - PEEK Lumbar Interbody Fusion Cage
Scope Start	Raw PEEK Pellet Receipt & Material Verification
Scope End	Final Dimensional Inspection & Lot Release
Device Classification	Class II - 510(k) (21 CFR 888.3070)
Standard Applied	AIAG-VDA FMEA 4th Edition 2019
Regulatory Framework	21 CFR 820 (QSR) | ISO 13485:2016 | ASTM F2026
Device Description	PEEK Lumbar Interbody Fusion Cage - Spinal Implant
Team Members	Mold Engineer, Quality Engineer, Process Engineer, Supplier Quality, R&D;
FMEA Facilitator	[Your Name] - Quality Engineer
FMEA Number	PFMEA-SPINE-001
Revision	Rev A
Date	2026-06-22
Next Review Date	6 months post-transfer validation
Status	Draft - Internal Review

AIAG-VDA FMEA 2019 | 21 CFR 820 | ISO 13485

2. Structure Analysis

STEP 2 - Structure Analysis | System Hierarchy Breakdown

System Level	Element Name	Function Contribution
Level 1 - System	Injection Molding Cell	Thermoplastic implant manufacturing environment
Level 2 - Subsystem	Injection Molding Press	Melts and injects PEEK at controlled temp/pressure
Level 3 - Component	Mold Tool (Cavity/Core)	Defines part geometry, surface finish, and tolerances
Level 3 - Component	Barrel / Screw Assembly	Plasticizes PEEK; controls melt temperature and shear
Level 3 - Component	Hot Runner / Gate System	Delivers melt uniformly; controls gate vestige height
Level 3 - Component	Material Handling & Drying	Ensures PEEK moisture <0.02% before processing
Level 3 - Component	Process Controller (HMI)	Enforces validated parameter recipe (IQ/OQ/PQ)
Level 4 - Interface	CMM / Vision Inspection Station	Dimensional and visual verification post-mold
Level 4 - Interface	Electronic Traveler (DHR)	Records lot, material CoA, operator, and process params

AIAG-VDA FMEA 2019 | 21 CFR 820 | ISO 13485

3. Function Analysis

STEP 3 - Function Analysis | Intended Functions & Requirements

Process Element	Intended Function	Requirement Satisfied
Mold Tool (Cavity/Core)	Form cage geometry to OD/ID and footprint tolerances per drawing; achieve Ra <= 1.6 um on endplates	Device drawing dimensional stack; osseointegration surface spec per ASTM F2026
Barrel / Screw Assembly	Maintain PEEK melt temperature 380-400 C; prevent thermal degradation of polymer	PEEK processing spec; material property retention per ISO 10993-1 biocompatibility
Hot Runner / Gate System	Deliver bubble-free melt shot; gate vestige height <= 0.3 mm above parting line	Device drawing gate vestige callout; visual inspection acceptance criteria
Material Handling & Drying	Dry PEEK pellets to moisture <= 0.02% at 150 C for >= 4 hours before molding	PEEK processing guideline; prevention of hydrolytic degradation and splay defects
Process Controller (HMI)	Lock and enforce validated parameter recipe; flag deviation >5% from setpoint	Validated manufacturing process (IQ/OQ/PQ); 21 CFR 820.70(i) software control
CMM / Vision Inspection	Verify footprint, height, endplate geometry, and gate vestige against print tolerance	Inspection plan IP-SPINE-002; acceptance criteria per device drawing Rev D
Electronic Traveler (DHR)	Gate operation sequence; capture material lot, CoA link, operator, and process params	21 CFR 820.184 Device History Record; ISO 13485 7.5.3 traceability

AIAG-VDA FMEA 2019 | 21 CFR 820 | ISO 13485

4. Failure Analysis

STEP 4 - Failure Analysis | Failure Modes, Effects & Causes

Process Step	Failure Mode	Effect on Patient / Downstream Process	Severity (1-10)	Root Cause	Current Prevention Control	Current Detection Control
Injection Molding - Parting Line	Flash / fin exceeding 0.1 mm at parting line	Sharp edge risk during implant handling; potential particulate in surgical field	6	Insufficient clamp force; worn parting line; mold damage	Clamp force setpoint in validated recipe; mold PM schedule	100% visual inspection per WI-VIS-003; AQL sampling CMM
Injection Molding - Geometry	Outer footprint or height OOS (dimensional deviation)	Implant-endplate mismatch; subsidence risk; non-union - patient harm	8	Mold wear / thermal expansion; incorrect recipe offset; shot-to-shot variation	Recipe lock in HMI; mold qualification per OQ; SPC on critical dims	CMM inspection per sample plan SP-SPINE-01; first-article 100%
Material Handling	Wrong PEEK grade loaded (incorrect molecular weight / filler)	Incorrect mechanical properties; implant fracture or fatigue failure in vivo	9	Manual material selection; similar pellet appearance; label misread	Material labeling SOP; operator training	Visual label check only - no barcode/CoA lock (DETECTION GAP)
Injection Molding - Fill	Short shot / incomplete mold fill	Structural porosity in load-bearing cage; fracture risk under spinal loading	7	Insufficient injection pressure; blocked gate; material viscosity variation	Injection pressure setpoint and transfer position in validated recipe	Visual inspection - porosity may be subsurface and missed (GAP)
Gate / Trim	Gate vestige height > 0.3 mm above parting surface	Mechanical interference with endplate; surgeon re-work; particulate generation	7	Gate freeze inconsistency; dwell time variation; manual trim variability	Dwell time in recipe; gate vestige spec on drawing	First-article caliper check; periodic sampling only
Injection Molding - Surface	Endplate surface texture OOS (Ra > 1.6 um or peak pattern missing)	Reduced bone in-growth; long-term instability and re-operation risk	8	Worn or damaged mold texture; contamination on cavity surface; improper mold release	Mold surface inspection at PM; release agent SOP	Profilometer sample check per lot - not 100% (DETECTION GAP)

AIAG-VDA FMEA 2019 | 21 CFR 820 | ISO 13485

5. Risk Rating

AIAG-VDA 2019 Action Priority (AP) -- NOT Traditional RPN

AIAG-VDA 2019 replaces RPN (SxOxD) with Action Priority (AP: High / Medium / Low). For spinal implants, an undetected material substitution (wrong PEEK grade) reaching a patient represents a catastrophic fracture risk regardless of how rarely it occurs. AP weights Detection gaps heavily - a D=7 with any S>=9 is automatically High priority.

AP Lookup: S9-10+D7-10->H | S9-10+D4-6+O>=4->H | S7-8+D7-10+O>=4->H | All others->M or L

STEP 5 - Risk Rating | AIAG-VDA Action Priority Table

Failure Mode	S (Severity 1-10)	O (Occurrence 1-10)	D (Detection 1-10)	Action Priority	AP Rationale	Legacy RPN (Reference Only)
Flash / fin at parting line	6	4	3	L	Lower severity with adequate controls	72
Outer footprint or height OOS	8	4	5	M	S=7-8 + D=4-6 + O>=4: moderate risk profile	160
Wrong PEEK grade loaded	9	3	7	H	S>=9 + D>=7: undetected critical failure - patient risk	189
Short shot / incomplete fill	7	3	2	L	S=7-8 + D<=3: detection adequate for severity level	42
Gate vestige height > 0.3 mm	7	4	4	M	S=7-8 + D=4-6 + O>=4: moderate risk profile	112
Endplate surface texture OOS	8	5	7	H	S=7-8 + D>=7 + O>=4: high-severity, frequent, poorly detected	280

AIAG-VDA FMEA 2019 | 21 CFR 820 | ISO 13485

6. Optimization

STEP 6 - Optimization / Corrective Actions | Risk Reduction Plan

Initial AP	Failure Mode	Action Description	Action Owner	Target Date	Revised S	Revised O	Revised D	Revised AP
H	Wrong PEEK grade loaded	Implement material barcode scan at press hopper: eDHR locks out start until scanned lot matches approved CoA on traveler. Certificate of Analysis auto-linked to DHR. Validate per 21 CFR 820.70(i); 100% operator training before go-live.	Quality Engineer	2026-Q3	9	2	2	M
H	Endplate surface texture OOS	Install inline profilometer at ejection station: 100% Ra measurement, SPC alert at Ra > 1.4 um. Mold surface inspection added to PM checklist at every 5,000 shots. Update Control Plan CP-SPINE-001 and WI-MOLD-002 Rev B.	Process Engineer	2026-Q3	8	3	3	L
M	Outer footprint or height OOS	Add 100% vision system check for critical footprint and height at ejection. Non-conforming parts automatically rejected to quarantine bin. SPC charts for cavity-specific dimensional trending.	Quality Engineer	2026-Q4	8	2	2	L
M	Gate vestige height > 0.3 mm	Add gate vestige height to mandatory first-article CMM program (currently omitted). Increase periodic sampling from 1/lot to 5/lot until Cpk >= 1.67 demonstrated over 30 lots.	Quality Engineer	2026-Q4	7	3	2	L

AIAG-VDA FMEA 2019 | 21 CFR 820 | ISO 13485

7. Results Summary

STEP 7 - Results & Documentation Summary

pFMEA Metrics Summary (auto-computed)

Total Failure Modes Analyzed	6
Initial High (H) Action Priority	2
High AP After Actions Implemented	0
Medium (M) AP - Initial	2
Medium (M) AP - After Actions	1
Low (L) AP - After Actions	5
Total Corrective Actions Assigned	4
Actions with Owner Assigned	4
Actions with Target Date	4

Linked Quality System Documents

Document Type	Document Number	Title / Description	Status
Control Plan	CP-SPINE-001 Rev A	Injection Molding - PEEK Spinal Cage Control Plan	In Review
Work Instruction	WI-MOLD-002 Rev B	Injection Molding Setup & Operation - PEEK Cage	Released
Inspection Plan	IP-SPINE-002 Rev A	CMM & Vision Inspection - Lumbar Interbody Cage	Released
IQ/OQ Protocol	IQ-MOLD-003	Injection Press Qualification - PEEK Processing Parameters	Pending
PQ Protocol	PQ-SPINE-CAGE-001	Process Validation - PEEK Cage Injection Molding Transfer	Planned
Risk Management	RM-SPINE-2024-02	Device Risk Management File (ISO 14971) - Fusion Cage	In Review
DFMEA	DFMEA-SPINE-001	Design FMEA - PEEK Lumbar Interbody Fusion Cage	Released
SOP	SOP-QE-022	pFMEA Creation and Maintenance Procedure	Released

Archive Location: [PLM System] -> Projects -> SPINE-CAGE-TRANSFER-2024 -> FMEA -> PFMEA-SPINE-001_RevA.xlsx

AIAG-VDA FMEA 2019 | 21 CFR 820 | ISO 13485